

How Fine-Grained Is Phenomenal Consciousness?

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1. Introduction

Central to the debate in the metaphysics of mind is a clash of intuitions about the nature of phenomenal consciousness:

1. Mental states are distinct from physical states.
2. There is causal interaction between the mental and the physical.
3. The universe is physically causally closed.

Upon brief inspection, it becomes immediately apparent that these intuitions are jointly incompatible. Accepting (3) commits you to the following: if a physical event e has a cause at time t , then e has a sufficient physical cause at t .¹ Given the explanatory completeness of contemporary physics with respect to physical phenomena in support of (3), and assuming no cases of genuine causal overdetermination, only one of (1) or (2) could be true at a given time. This is because, if all physical events have a sufficient physical cause, a mental event can only enter into the causal story of the universe if it can be reduced down to, or identified with, some physical event. But that is clearly in tension with what (1) says!²

At the dawn of the philosophy of mind, Descartes' substance dualism tried to permit causal interaction between immaterial souls and our material bodies, thus violating physical causal closure. Meanwhile, Malebranche's occasionalism famously denied any causal connection between mind and matter whatsoever, asserting instead a preestablished harmony to explain the correlations between mind and matter, with God having monopolized the world's causal power. However, beginning at the turn of the 20th century, most philosophers of mind have come to see these moves as metaphysically extravagant, and looked to (1) as the intuition to reject. Gilbert Ryle and the behaviorists began that project by identifying mental states with behavioral dispositions, but they seemed to exclude the qualitative character of mentality from their theory of mind. JJC Smart sought to identify and integrate the place of mind in the physical world, but his identity theory failed to account for the multiple realizability of consciousness. Finally, Hilary Putnam and the functionalists found a way to reconcile the physical basis of phenomenal consciousness with multiple realizability by identifying mental types with functional roles rather than any specific physical realizer. And thus, the metaphysics of mind was taken to be naturalized, and mind identified with matter.

This optimistic story of physicalism, however, would not stay for long. Frank Jackson and Mary, the colorblind scientist, would emphasize the idea that our physical theories fail to explain why

¹ However, this is not an uncontentious principle. See Papineau (2001) for a defense of physical causal closure.

² Of course, there are different ways to spell out what distinctness means when it comes to this intuition. Is it that mind is ontologically different from matter, or it is only that by some mechanism we misrepresent phenomenal consciousness to have some properties that give it the illusion of distinctness? Depending on what distinctness means, (1) may or may not be compatible with (2) and (3). See Chalmers (2002) and Frankish (2016) for contrasting positions on the topic.

experience arises, while David Chalmers and his zombies would put pressure on the idea that the correct spatial arrangement of physical matter satisfying some functional role in a system would entail the existence of experience. But perhaps the most damning of all, for any standard physicalist theory of mind, one seeks a general mechanism to accommodate, or thereby do away with, what's come to be known as the explanatory gap³. Put succinctly, the physicalist must explain how it is the case that the particular spatial arrangements of some physical state gives rise to the subjective, qualitative content of phenomenal consciousness. Indeed, it seems totally mysterious to us under the physicalist worldview why, say, the firing of my C-fibers would perfectly correlate with my subjective experience of pain. What plausible mechanism is there, under physicalism, that can deliver this conclusion while also explaining away the possible connection between my C-fibers firing and my sensation of joy? Why is there a connection between my physical states and my mental states at all? Ultimately, if physicalism about phenomenal consciousness is correct, then a complete theory of the mind should posit some guiding principle to either successfully bridge the explanatory gap between mind and body, or dissolve the gap itself. The aim of physicalists today, therefore, is to provide this guiding principle.

2. Ground Functionalism

Jonathan Schaffer has recently attempted to provide such a principle in his (2021) paper, "Ground Functionalism," by postulating the existence of a mind-making operation that maps the functional state of some system to some mental property. The theoretical upshot of this move, for Schaffer, is to provide a viable metaphysical explanation of the relationship between mind and matter via a linking grounding principle, thus bridging the explanatory gap. Schaffer first presents us with some illustrative examples of this kind of move in less controversial cases to see how such a grounding principle works. Consider the relationship between Socrates and the singleton set containing only Socrates. Following grounding orthodoxy, we say the existence of Socrates grounds the existence of the singleton set {Socrates}, despite the fact Socrates and {Socrates} are necessarily coextensive—while the existence of Socrates explains why the singleton set exists, the converse explanation fails to hold. Grounding, as a relation, is often taken to be asymmetric, irreflexive, transitive, constitutive, and hyperintensional. Thus, grounding is best understood as a notion of metaphysical dependency which specifies which facts obtain in virtue of which other facts. The examples Schaffer invokes can be used to understand a systemic generalization of this idea.⁴ Consider again, Socrates and the singleton set containing Socrates, or any entity and the singleton set containing only that entity. Is there a viable metaphysical explanation of the relationship between an entity and its singleton set? Schaffer answers in the affirmative: {Socrates} exists in virtue of the existence of Socrates and a *set formation principle*. It is this set formation principle that specifies how, for any given individual, a corresponding singleton set is constructed. The grounding explanation Schaffer offers thus appeals to an operation that takes certain inputs as metaphysically prior and yields a derivative entity as its output. As Schaffer describes it,

"We can think of set formation as a machine that clamps things into sets. For every plurality of inputs one feeds into this machine, one gets out a set with all and only those inputs as members. So if one

³ See Levine (1983) for the classic formulation of this challenge.

⁴ While Schaffer discusses the prevalence of grounding as a viable metaphysical explanation across a variety of phenomena (including mereology, metaethics, and quantum mechanics), for the sake of brevity, I will only focus on the case of set formation.

feeds Socrates into this machine, one gets out {Socrates}... there is a viable metaphysical explanation for why {Socrates} exists, from the existence of Socrates, via the principle of set formation. After all, given that Socrates exists and set formation is at work, it is no mere accident that {Socrates} exists too. For there is a grounding principle: set formation, which clamps every plurality of inputs into a so-membered set (never mind how set formation came to operate—we are just assuming it is at work). And there is an input: Socrates (never mind where Socrates came from—we are just assuming he is there). {Socrates} is the output of this principle on this input, and so {Socrates} exists.”

Note, the explanation in this case is neither reductive nor identity-based—rather the singleton set containing Socrates exists *in virtue of* Socrates plus the set-formation principle as a matter of grounded determination. If grounding can systematically explain the dependence of higher-level entities on lower-level ones via generative principles in the case of set formation (and mereological fusion, right-making, and particle enactment), then it is natural to ask whether a similar strategy can be extended to the mental case. By evaluating and defending the legitimacy of grounding-backed explanations across a variety of worldly phenomena, Schaffer hopes to generalize the theoretical role of grounding as a metaphysically generative operation to explain the dependency relation between mind and body. The methodology here is explicitly abductive—Schaffer’s proposal to posit a mind-making operation is motivated by inference to the best explanation. In doing so, Schaffer hopes to bridge the explanatory gap.

It’s important to consider why the explanatory gap seems so insurmountable for the physicalist, and how Schaffer’s approach does away with it. Namely, standard presentations of the gap are motivated by the constraint of *epistemic transparency*. Put more succinctly, transparency is a deductive constraint, with a demand that the link between mind and body be “evident to pure reason, *a priori* knowable, or inconceivable otherwise.”⁵ But Schaffer points out that transparency fails to obtain in paradigmatic cases of grounding! It seems conceivable that the set formation principle fails to link Socrates to the singleton set {Socrates}.⁶ Is there then an explanatory gap between sets and their constituents? Additionally, Schaffer holds that transparency fails for other generative enterprises, such as causality and the laws of nature, so why would it hold in the case of grounding? The successful grounding principles Schaffer has posited up to this point have explained metaphysical dependency between two realms of entities by way of generative operations, not by eliminating explanatory opacity. By parity of reasoning, Schaffer argues the lack of transparency in the mind-body case cannot motivate a rejection of physicalism any more than the lack of transparency in the set-formation case can by itself motivate a rejection of the grounding of sets in their constituents. It is via this process of abductive inference that Schaffer posits his mind-making operation as the best explanation for the systemic correlations between the physical realizers in a system that enact some functional role and mental properties. In this way, Schaffer aims to give a promising

⁵ Schaffer goes on to specify the transparency constraint, writing “For instance one might require an *a priori* entailment from the grounds to the grounded, so that an ideal reasoner in her armchair, given the information about the grounds and full conceptual competence, could deduce the facts about the grounded on that basis (Chalmers & Jackson 2001: 351). Or—perhaps equivalently—that it should be inconceivable for the link to fail, so that an ideal reasoner would regard any other view as contradictory (Chalmers 1996: 73–76).” But given nothing important hinges on how we spell out transparency, we do not have to take a stance of which is the correct reading.

⁶ And likewise in other cases of grounding. It’s conceivable that mereological fusion fails to link simples to ordinary objects (in the case of a nihilist universe), the right-making principle to link desires to moral properties (assuming a different theory of moral value), etc.

solution to the explanatory gap by embedding mentality within the broader grounding structure of the world.

3. Setting an Upper Bound

As it happens, there is a tension in the principles Schaffer's theory of ground functionalism relies on, along with some background logical assumptions. Namely, if phenomenal consciousness is fine-grained enough to mirror the structure of logical syntax, given some orthodox views about the nature of ground, the theory of ground functionalism is rendered untenable. To show this conclusion, I will first formalize plausible readings of the principles Schaffer relies upon in the language of higher-order logic, along with associated principles motivated elsewhere in the grounding literature, before going on to derive my intended results. The upshot of the arguments I provide is to show that attempts to describe physicalism in terms of grounding, along with a higher-order comprehension schema, lead to inconsistency. Hence, the grain of our mental properties must be more coarse than the grain of our logical syntax.

I will begin by introducing the primitive predicate identifications and relations I will be relying on in my argument. First, let $\text{Mental}(M)$ represent a univariate predicate indicating that M is a mental property. Next, let $P(s)$ indicate that property P is instantiated in system s . Finally, let the operator $<$ represent the relation of metaphysical grounding, such that $P(s) < Q(s)$ indicates that $P(s)$ grounds $Q(s)$, or $Q(s)$ obtains in virtue of $P(s)$.

Now, we can formalize the distinctive core commitment of Schaffer's theory of mind: the postulation of a mind-making operation. As elucidated above, mind-making is a metaphysically generative grounding operation that takes physical realizers as inputs and links them to mental properties as outputs. Informally, Schaffer is committed to the idea that if a property is a mental property, then it must be the case that for any system that instantiates the mental property, there exists some physical base also instantiated in the system that grounds it. We can formalize this mind-making principle as follows:

$$\text{(Mind-Making): } \forall M(\text{Mental}(M) \rightarrow \forall s(M(s) \leftrightarrow \exists P(P(s) \wedge P(s) < M(s)))$$

The second principle I will be using is a standard background assumption in the grounding literature: factivity.⁷ Informally, factivity says that if P grounds Q , then P and Q must both actually obtain. Schaffer seems to endorse this principle within his broader metaphysical framework, writing, "the metaphysician with the ambition to explain how and why the world is as it is should offer theories of what grounds what".⁸ This suggests that, for Schaffer, grounding is a relation between domains that is intended to provide explanatory insight for phenomena we encounter. Since explanations cannot hold between non-existent facts, grounding relations can't hold between facts that don't obtain or uninstantiated properties. This can be spelled out formally below:

$$\text{(Factivity): } \forall M, P, s(P(s) < M(s) \rightarrow (P(s) \wedge M(s)))$$

⁷ See Fine (2012) for the seminal defense of factivity.

⁸ See Schaffer (2009), where he goes on to further defend this idea.

Finally, the last principle I will be relying on is comprehension, which allows us to introduce the existence of a property corresponding to specific formulas. Crucially, for our purposes, comprehension allows us to introduce the existence of a property corresponding to any well-formed condition on properties and systems. In particular, it guarantees that for any formula $\phi(x)$ specifying a condition on instantiation, there exists a property whose instances are exactly those systems satisfying ϕ . This will allow us to define a property by reference to its grounding conditions, and then reason about whether such a property can be instantiated under Schaffer's principles. It can be specified formally below:

(Comprehension): $\exists P \forall x (P(x) \leftrightarrow \phi(x))$

From here, the argument can take off by constructing a property defined as the property that obtains if the physical states that ground it are inactive. Formally, the property is defined as below:

Define $\Phi := \lambda Q \lambda s. \forall P ((P(s) \prec Q(s)) \rightarrow \neg P(s))$

From comprehension, we can infer the existence of a property, M^* , that corresponds to the property defined above. Now, we face a dilemma regarding the status of this property: is M^* instantiated in some system s ? For reductio, let's say the answer is no. If M^* does not obtain in system s , then by the definition of M^* , the condition for its instantiation is false. Negating the universal condition implies the existence of a counter-example: there must be at least one physical property—let us call it K —that acts as a ground for M^* and is currently active in system s . However, by factivity, if K is active and grounds M^* , then the grounded fact M^* must itself obtain. Thus, from the assumption that M^* does not obtain in s , now we have reached a contradiction: M^* is instantiated in s and it is not. Formally, the argument looks as such:

1. Define $\Phi := \lambda Q \lambda s. \forall P ((P(s) \prec Q(s)) \rightarrow \neg P(s))$
2. From 1, Comprehension: $\exists M^* \forall s (M^*(s) \leftrightarrow \forall P (P(s) \prec M^*(s) \rightarrow \neg P(s)))$
3. Assume $\neg M^*(s)$
4. From 2: $\neg \forall P (P(s) \prec M^*(s) \rightarrow \neg P(s))$
5. From 4: $\exists P \neg (P(s) \prec M^*(s) \rightarrow \neg P(s))$
6. From 5: $\exists P (P(s) \prec M^*(s) \wedge P(s))$
7. From 6, EI: $K(s) \prec M^*(s) \wedge K(s)$
8. From 7, Factivity: $K(s) \wedge M^*(s)$
9. From 3, 8: $M^*(s) \wedge \neg M^*(s)$

Since assuming M^* does not hold of s gives rise to contradiction, we must therefore accept that M^* does hold. Now, we can ask a question about the ontological status of M^* : is it mental, or is it not? Assume, for reductio again, that M^* is mental. But, given Mind-Making, this too cannot be the case. The reasoning is this: if M^* is mental, then the antecedent condition of Schaffer's mind-making principle will hold, thus establishing some physical realizer that is active in s and grounds M^* . Thus, we can infer the existence of some physical property, K , that is active in s and stands in a grounding relation to M^* . However, by checking this result against the definition of M^* itself, we reach a problem: the property M^* is defined so that it obtains if and only if every physical realizer grounding it is inactive. Applying this

definition to our specific realizer K , we reach contradiction: the physical ground for M^* is both instantiated and not instantiated in s . The argument can be presented formally as such:

1. Define $\Phi := \lambda Q \lambda s. \forall P((P(s) < Q(s)) \rightarrow \neg P(s))$
2. From 1, Comprehension: $\exists M^* \forall s(M^*(s) \leftrightarrow \forall P(P(s) < M^*(s) \rightarrow \neg P(s)))$
3. Assume $M^*(s) \wedge \text{Mental}(M^*)$
4. From 3, Mind-Making: $\exists P(P(s) \wedge P(s) < M^*(s))$
5. From 4, EI: $K(s) \wedge K(s) < M^*(s)$
6. From 2, EI, UI: $K(s) < M^*(s) \rightarrow \neg K(s)$
7. From 5, 6: $K(s) \wedge \neg K(s)$

Thus, we have reached a tension in Schaffer's Ground Functionalism: M^* must be an instantiated non-mental property in s for his theory to remain consistent. Why then, is this a problem for Schaffer? Well, if comprehension guarantees the existence of property M^* , and if we accept a stronger reading of Ground Physicalism, where all instantiated properties are grounded in some physical base, then any attempt to satisfy Ground Physicalism by positing an active physical ground for M^* will run into inconsistency once more. The principle I will rely on for this derivation looks formally as follows:

(Ground Physicalism): $\forall M \forall s(M(s) \rightarrow \exists P(P(s) \wedge P(s) < M(s)))$

The challenge is simple and follows by reasoning analogous to Russell's paradox: by definition, M^* obtains in a system if and only if its physical grounds are absent. We have seen how contradiction will arise if we deny that M^* is instantiated in s or if we accept it is instantiated and is a mental property. But now, if we say M^* is instantiated in s and is not mental, and if we attempt to posit some active physical ground for M^* in order to satisfy Ground Physicalism, then by definition, M^* immediately entails that this ground is inactive. Formally, the derivation proceeds as follows:

1. Define $\Phi := \lambda Q \lambda s. \forall P((P(s) < Q(s)) \rightarrow \neg P(s))$
2. From 1, Comprehension: $\exists M^* \forall s(M^*(s) \leftrightarrow \forall P(P(s) < M^*(s) \rightarrow \neg P(s)))$
3. From above: $M^*(s) \wedge \neg \text{Mental}(M^*)$
4. From 3, Ground Physicalism, UI, EI: $P(s) \wedge P(s) < M^*(s)$
5. From 2, UI, EI: $P(s) < M^*(s) \rightarrow \neg P(s)$
6. From 5, 4: $P(s) \wedge \neg P(s)$

I have shown that a variety of contradictions arise if we accept Mind-Making, Factivity, Comprehension, and Ground Physicalism. Thus, one of the principles must be rejected. Upon inspection, it seems the most viable candidate for rejection here is Comprehension. This is because Factivity stands on solid footing, and is supported widely by grounding theorists, and Schaffer cannot give up Mind-Making or Ground Physicalism without conceding the central theses of his worldview. This leaves Comprehension as the culprit. By rejecting the unrestricted comprehension principle that validates the existence of any property that corresponds to a well-formed formula, we can thereby block the existence of the problematic property M^* . We can thus avoid the contradictions generated by denying that the formulation of a logical predicate is coextensive with the existence of a corresponding property in the

world. The metaphysical upshot of these arguments is to show that the grain of phenomenal consciousness must be coarser than the grain of our logical syntax.

4. Setting a Lower Bound

Having set an upper bound on how finely mental properties are to be individuated, I will now develop an argument providing a lower bound to the grain of phenomenal consciousness. Here, I will not respond to any particular theory of mentality. Instead, I will argue that the view that makes identifications on the basis of necessary coextensiveness—known as intensionalism—cannot be maintained given widely accepted commitments about mind-body supervenience and the nature of mental explanation. Specifically, I will argue that if we accept these very plausible principles about the mind, intensional individuation collapses the explanatory distinction between mental properties and physical properties, yielding a case of self-explanatory phenomena. The upshot of this argument is that intensionalism sets the bar for property individuation at too coarse a level—mental properties must be more fine-grained than sets of worlds.

I will begin again by postulating the primitive predicate identifications that I will rely on in my argument. As before, let $\text{Mental}(M)$ represent a univariate predicate that indicates M is a mental property. Additionally, let $\text{Expl}(p, q)$ be a bivariate predicate that indicates p explains q . Finally, let quantification over variable P indicate quantification over physical properties.

Now, we can formalize two commonly held commitments for all physicalists. While mind-body supervenience has been spelled out in a variety of ways by philosophers who wanted varying degrees of modal strength (e.g. local vs global supervenience),⁹ for our purposes, we will rely on a modally strengthened supervenience principle generally endorsed by physicalists. Formally, it looks as follows:

(Mind-Body Supervenience): $\forall M(\text{Mental}(M) \rightarrow \exists P \Box(P \leftrightarrow M))$

In other words, for all mental properties, there exists some physical property that necessarily covaries with that mental property. Note here, I am not assuming that the physical property must be the supervenient base, and that the mental property supervenes upon it. Thus, I hope Russellian Monists and panpsychists who accept some modal relation at least as strong as supervenience between the intrinsic, phenomenal properties of some individual and its physical, structural properties will endorse this principle as well.¹⁰

The second principle of mind is that of explanatory asymmetry. Intuitively, it seems the direction of explanation goes from the physical domain to the mental one—that is, facts about mental states obtain in virtue of facts about their physical realizers, and not vice versa. This principle can be formalized as follows:

(Physical Explanation of Mental): $\forall M(\text{Mental}(M) \rightarrow \exists P(\text{Expl}(P, M)))$

⁹ On this point, see Kim (1984)

¹⁰ See Strawson (2017) and Mørch (forthcoming) as panpsychists who would endorse this formulation of mind-body supervenience.

This is the principle that will separate those who are committed to physicalism from those who are not. The reason is this: explanation is an asymmetric relation that holds between two states of affairs, one of which plays the role of explanans, and the other of which occupies the role of explanandum.¹¹ Evaluating the spectrum of views when it comes to the mind-body problem, it becomes clear only the physicalist would accept that physical facts stand in the correct role of explanans with respect to the mental—the dualist would deny that mental facts admit of physical explanations at all, and the panpsychist would reverse the direction of explanation by treating the mental as metaphysically fundamentally. Hence, a commitment to Physical Explanation of Mental marks a substantive and distinctive thesis of physicalism that demarcates it from the other views in the philosophy of mind.

Before I make the argument against intensionalism, let us also formalize what intensionalism says first, and a generally accepted principle of no self-explanation.

(Intensionalism): $\forall Q \forall M (\Box(Q \leftrightarrow M) \rightarrow Q = M)$

(No Self-Explanation): $\forall Q (\neg \text{Expl}(Q, Q))$

From here, the argument can be advanced as follows: first, we assume the existence of some arbitrary mental property. For demonstrative purposes, let us assume the existence of the property of being in pain. From Mind-Body Supervenience, we can infer the existence of some physical property that necessarily covaries with the property of being in pain. As philosophers have long used as a toy example, let us say this is the property of C-fibers firing. And from Physical Explanation, we can infer that the direction of explanation goes from physical to mental, with the property of having one's C-fibers firing explaining the property of being in pain. Now, given intensionalism about mentality, we can say that the property of having one's C-fibers firing is identical to the property of being in pain. But given the intersubstitutability of identities, at least in non-opaque contexts, we can now say that the property of being in pain explains the property of being in pain. But now this is a clearly objectionable case of self-explanation! Formally, the argument looks as follows:

1. Assume arbitrary mental property: $\text{Mental}(M)$
2. From 1, Supervenience, EI: $\Box(P \leftrightarrow M)$
3. From 1, Physical Explanation, EI: $\text{Expl}(P, M)$
4. From 2 and Intensionalism: $P = M$
5. From 3 and 4: $\text{Expl}(M, M)$
6. From 5 and No Self-Explanation, UI: $\text{Expl}(M, M) \wedge \neg \text{Expl}(M, M)$

Thus, given two very plausible principles about the nature of the mind, along with a constraint on explanation and intensionalism, we are led to contradiction. Since it seems all non-grain principles stand on solid footing, if we are to avoid this conclusion, intensionalism must be the principle to reject. Hence, the lower bound for phenomenal consciousness' grain must sit at a level higher than intensionalism.

¹¹ It also seems a plausible constraint that if some fact A fully explains some other fact B , then A is at least as fundamental as B , with explanation being an asymmetric guide to fundamentality. See Bennett (2017) and Raven (2015) for a defense.

5. Conclusion

The challenge of individuating mentality and making the proper identifications between the mental and the physical under any physicalist theory is an important and open question in the metaphysics of mind that will inevitably bear on questions now being addressed by metaphysicians working in a higher-order setting. I've attempted here to set bounds on how fine phenomenal consciousness can be, to being more fine-grained than sets of worlds and more coarse-grained than logical syntax, but I suspect other questions that preoccupy those metaphysicians working on consciousness will find novel insight from the developments of higher-order metaphysics. For instance: in the latter half of the 20th century, Jaegwon Kim formulated his now well-established causal exclusion argument against non-reductive forms of physicalism by showing a tension between mind-body supervenience, physical causal closure, the causal efficacy of the mental, and the irreducibility of mind to matter.¹² Since then, reductive physicalists following in Kim's footsteps have advanced a bold position on the nature of mental causation: our minds are such that every mental event is a physical event, or it is an epiphenomenon.¹³ But even if Kim's causal exclusion works to show it is genuinely the case that non-reductive forms of physicalism can't account for the causal efficacy of the mental, that does not thereby automatically grant the reductive physicalist an exemption from providing an explanation of mental causation under the reductive picture. This is because, even if some mental event is necessarily equivalent to some physical event, it does not automatically follow that the mental event plays some relevant role in the causal story of some phenomena occurring, even if the physical event it is equivalent to does! The mere fact that, say, my C-fibers firing causes my crying out does not automatically warrant us to say that my feeling of pain causes my crying out even if my C-fibers firing is identical to my feeling of pain, for there could be a hyperintensional distinction drawn between mind and body. Now, whether or not mentality is a hyperintensional phenomenon remains an open question for those working in the metaphysics of mind. But it is at least not obvious to me that phenomenal consciousness is a phenomenon immune from it. Thus, I hope to have drawn interest to a new research project focused on those questions found within the metaphysics of mind that can utilize the rigor and formalism of recent developments in the higher-order metaphysics literature to better investigate the ultimate nature of phenomenal consciousness.

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¹² This position has been canonically defended in Kim (2005)

¹³ See for example McLaughlin (2015), who goes on to defend this position

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